

The computations of this section were done on the Team Ouragan (INRIA, IMJ-PRG) servers with 128GB of RAM and 32 Intel i9-13900 processors. The code was run using SageMath version 9.5. We may add further examples in the future.

Example 1. $D = 12$ and $\ell = 3$.

Table 1: Minimal polynomial for $D = 12$, $\ell = 3$

p	$P_{u_\ell(\omega)}$	p	$P_{u_\ell(\omega)}$
5	$x^2 + \frac{8}{5}x + 1$	7	$x^2 + \frac{13}{7}x + 1$
17	$x^2 - \frac{16}{17}x + 1$	19	$x^2 - \frac{11}{19}x + 1$
29	$x^2 + \frac{40}{29}x + 1$	31	$x^2 + \frac{46}{31}x + 1$
41	$x^2 - \frac{80}{41}x + 1$	43	$x^2 - \frac{22}{43}x + 1$
53	$x^2 - \frac{56}{53}x + 1$	67	$x^2 + \frac{109}{67}x + 1$
79	$x^2 + \frac{131}{79}x + 1$	89	$x^2 - \frac{160}{89}x + 1$
101	$x^2 - \frac{40}{101}x + 1$	103	$x^2 - \frac{37}{103}x + 1$
113	$x^2 - \frac{224}{113}x + 1$	127	$x^2 - \frac{146}{127}x + 1$
137	$x^2 - \frac{176}{137}x + 1$	139	$x^2 - \frac{251}{139}x + 1$
149	$x^2 + \frac{280}{149}x + 1$	151	$x^2 - \frac{59}{151}x + 1$
163	$x^2 - \frac{299}{163}x + 1$	173	$x^2 - \frac{104}{173}x + 1$
197	$x^2 + \frac{56}{197}x + 1$	199	$x^2 - \frac{277}{199}x + 1$
211	$x^2 + \frac{253}{211}x + 1$	223	$x^2 - \frac{338}{223}x + 1$
233	$x^2 - \frac{416}{233}x + 1$	257	$x^2 + \frac{64}{257}x + 1$
269	$x^2 - \frac{520}{269}x + 1$	271	$x^2 - \frac{299}{271}x + 1$
281	$x^2 - \frac{320}{281}x + 1$	283	$x^2 - \frac{458}{283}x + 1$
293	$x^2 - \frac{136}{293}x + 1$	307	$x^2 + \frac{358}{307}x + 1$
317	$x^2 - \frac{616}{317}x + 1$	331	$x^2 - \frac{661}{331}x + 1$
353	$x^2 + \frac{544}{353}x + 1$	367	$x^2 + \frac{491}{367}x + 1$
379	$x^2 - \frac{83}{379}x + 1$	389	$x^2 - \frac{680}{389}x + 1$
401	$x^2 + \frac{80}{401}x + 1$	439	$x^2 + \frac{94}{439}x + 1$

449	$x^2 - \frac{560}{449}x + 1$	461	$x^2 + \frac{760}{461}x + 1$
463	$x^2 - \frac{397}{463}x + 1$	487	$x^2 - \frac{349}{487}x + 1$
499	$x^2 + \frac{26}{499}x + 1$	509	$x^2 - \frac{440}{509}x + 1$
521	$x^2 + \frac{880}{521}x + 1$	523	$x^2 - \frac{803}{523}x + 1$
547	$x^2 + \frac{1093}{547}x + 1$	557	$x^2 + \frac{1064}{557}x + 1$
569	$x^2 - \frac{1040}{569}x + 1$	571	$x^2 + \frac{181}{571}x + 1$
593	$x^2 + \frac{736}{593}x + 1$	607	$x^2 + \frac{1187}{607}x + 1$
617	$x^2 - \frac{1216}{617}x + 1$	619	$x^2 - \frac{949}{619}x + 1$
631	$x^2 - \frac{587}{631}x + 1$	641	$x^2 + \frac{400}{641}x + 1$
643	$x^2 + \frac{314}{643}x + 1$	653	$x^2 - \frac{1144}{653}x + 1$
677	$x^2 + \frac{104}{677}x + 1$	691	$x^2 + \frac{1318}{691}x + 1$
701	$x^2 + \frac{520}{701}x + 1$	727	$x^2 - \frac{482}{727}x + 1$
739	$x^2 - \frac{1222}{739}x + 1$	751	$x^2 + \frac{179}{751}x + 1$
761	$x^2 - \frac{1520}{761}x + 1$	773	$x^2 + \frac{1496}{773}x + 1$
787	$x^2 - \frac{613}{787}x + 1$	797	$x^2 - \frac{1144}{797}x + 1$
809	$x^2 + \frac{560}{809}x + 1$	811	$x^2 - \frac{1514}{811}x + 1$
821	$x^2 - \frac{1400}{821}x + 1$	823	$x^2 - \frac{1621}{823}x + 1$
857	$x^2 - \frac{464}{857}x + 1$	859	$x^2 - \frac{1549}{859}x + 1$
881	$x^2 + \frac{1600}{881}x + 1$	883	$x^2 - \frac{443}{883}x + 1$
907	$x^2 - \frac{1453}{907}x + 1$	919	$x^2 - \frac{866}{919}x + 1$
929	$x^2 - \frac{1840}{929}x + 1$	941	$x^2 + \frac{1160}{941}x + 1$
953	$x^2 - \frac{1456}{953}x + 1$	967	$x^2 + \frac{253}{967}x + 1$
977	$x^2 - \frac{496}{977}x + 1$	991	$x^2 - \frac{1739}{991}x + 1$
1013	$x^2 - \frac{2024}{1013}x + 1$	1039	$x^2 + \frac{1403}{1039}x + 1$
1049	$x^2 + \frac{640}{1049}x + 1$	1051	$x^2 - \frac{1994}{1051}x + 1$

1061	$x^2 - \frac{1240}{1061}x + 1$	1063	$x^2 + \frac{2099}{1063}x + 1$
1087	$x^2 + \frac{851}{1087}x + 1$	1097	$x^2 + \frac{1856}{1097}x + 1$
1109	$x^2 + \frac{2200}{1109}x + 1$	1123	$x^2 - \frac{1021}{1123}x + 1$
1171	$x^2 + \frac{2221}{1171}x + 1$	1181	$x^2 - \frac{680}{1181}x + 1$
1229	$x^2 - \frac{280}{1229}x + 1$	1231	$x^2 - \frac{2101}{1231}x + 1$
1277	$x^2 - \frac{1496}{1277}x + 1$	1279	$x^2 - \frac{709}{1279}x + 1$
1289	$x^2 + \frac{1120}{1289}x + 1$	1291	$x^2 - \frac{1907}{1291}x + 1$
1301	$x^2 + \frac{2600}{1301}x + 1$	1303	$x^2 - \frac{419}{1303}x + 1$
1327	$x^2 - \frac{2638}{1327}x + 1$	1361	$x^2 - \frac{2480}{1361}x + 1$
1373	$x^2 + \frac{296}{1373}x + 1$	1399	$x^2 - \frac{1826}{1399}x + 1$
1409	$x^2 + \frac{2800}{1409}x + 1$	1423	$x^2 + \frac{2446}{1423}x + 1$
1433	$x^2 - \frac{1184}{1433}x + 1$	1447	$x^2 - \frac{1669}{1447}x + 1$
1459	$x^2 - \frac{218}{1459}x + 1$	1471	$x^2 - \frac{2834}{1471}x + 1$
1481	$x^2 + \frac{2240}{1481}x + 1$	1483	$x^2 + \frac{1597}{1483}x + 1$
1493	$x^2 - \frac{1064}{1493}x + 1$	1531	$x^2 - \frac{3013}{1531}x + 1$
1543	$x^2 + \frac{2843}{1543}x + 1$	1553	$x^2 - \frac{2944}{1553}x + 1$
1567	$x^2 - \frac{3107}{1567}x + 1$	1579	$x^2 - \frac{2134}{1579}x + 1$

Example 2. $D = 60$ and $\ell = 3$.

Table 2: Minimal polynomial for $D = 60$, $\ell = 3$

p	P
13	$x^4 + \frac{4070}{13^4}x^3 + \frac{6939843}{13^6}x^2 + \frac{4070}{13^4}x + 1$
19	$x^4 - \frac{144958}{19^4}x^3 + \frac{95336763}{19^6}x^2 - \frac{144958}{19^4}x + 1$
23	$x^4 + \frac{644776}{23^4}x^3 + \frac{460440114}{23^6}x^2 + \frac{644776}{23^4}x + 1$
29	$x^4 + \frac{812240}{29^4}x^3 + \frac{515763762}{29^6}x^2 + \frac{812240}{29^4}x + 1$
31	$x^4 - \frac{264868}{31^4}x^3 + \frac{212317638}{31^6}x^2 - \frac{264868}{31^4}x + 1$
37	$x^4 + \frac{2608970}{37^4}x^3 + \frac{2089928931}{37^6}x^2 + \frac{2608970}{37^4}x + 1$
41	$x^4 - \frac{6730720}{41^4}x^3 + \frac{13147290402}{41^6}x^2 - \frac{6730720}{41^4}x + 1$
47	$x^4 + \frac{360136}{47^4}x^3 - \frac{13374234606}{47^6}x^2 + \frac{360136}{47^4}x + 1$
73	$x^4 - \frac{41474290}{73^4}x^3 + \frac{147346701891}{73^6}x^2 - \frac{41474290}{73^4}x + 1$
79	$x^4 - \frac{117069722}{79^4}x^3 + \frac{980879199243}{79^6}x^2 - \frac{117069722}{79^4}x + 1$
83	$x^4 - \frac{35685496}{83^4}x^3 - \frac{159288038526}{83^6}x^2 - \frac{35685496}{83^4}x + 1$
89	$x^4 + \frac{41736640}{89^4}x^3 + \frac{357560932002}{89^6}x^2 + \frac{41736640}{89^4}x + 1$
97	$x^4 - \frac{245909170}{97^4}x^3 + \frac{2971334808051}{97^6}x^2 - \frac{245909170}{97^4}x + 1$
101	$x^4 - \frac{141534640}{101^4}x^3 + \frac{771673121682}{101^6}x^2 - \frac{141534640}{101^4}x + 1$
107	$x^4 + \frac{303770984}{107^4}x^3 + \frac{5015372696034}{107^6}x^2 + \frac{303770984}{107^4}x + 1$
139	$x^4 - \frac{702403922}{139^4}x^3 + \frac{16924405747803}{139^6}x^2 - \frac{702403922}{139^4}x + 1$
149	$x^4 + \frac{918278480}{149^4}x^3 + \frac{27117890859282}{149^6}x^2 + \frac{918278480}{149^4}x + 1$
151	$x^4 + \frac{165119878}{151^4}x^3 - \frac{14065704355557}{151^6}x^2 + \frac{165119878}{151^4}x + 1$
157	$x^4 + \frac{903642820}{157^4}x^3 + \frac{37152516427926}{157^6}x^2 + \frac{903642820}{157^4}x + 1$
167	$x^4 + \frac{1186902376}{167^4}x^3 + \frac{26584898411634}{167^6}x^2 + \frac{1186902376}{167^4}x + 1$
193	$x^4 - \frac{2055328990}{193^4}x^3 + \frac{131588777818851}{193^6}x^2 - \frac{2055328990}{193^4}x + 1$
199	$x^4 - \frac{3879866426}{199^4}x^3 + \frac{192310453856571}{199^6}x^2 - \frac{3879866426}{199^4}x + 1$

211	$x^4 + \frac{230705134}{211^4}x^3 - \frac{48786874479669}{211^6}x^2 + \frac{230705134}{211^4}x + 1$
227	$x^4 - \frac{6517845224}{227^4}x^3 + \frac{398073871170114}{227^6}x^2 - \frac{6517845224}{227^4}x + 1$
263	$x^4 - \frac{1088170424}{263^4}x^3 - \frac{261289857784206}{263^6}x^2 - \frac{1088170424}{263^4}x + 1$
269	$x^4 - \frac{1053543920}{269^4}x^3 + \frac{18446765341362}{269^6}x^2 - \frac{1053543920}{269^4}x + 1$
271	$x^4 - \frac{1729560118}{271^4}x^3 - \frac{276568167620037}{271^6}x^2 - \frac{1729560118}{271^4}x + 1$
277	$x^4 - \frac{4974673220}{277^4}x^3 + \frac{846850815822966}{277^6}x^2 - \frac{4974673220}{277^4}x + 1$
281	$x^4 + \frac{17988446080}{281^4}x^3 + \frac{1956731128025442}{281^6}x^2 + \frac{17988446080}{281^4}x + 1$
313	$x^4 + \frac{8727639650}{313^4}x^3 + \frac{1775441952045843}{313^6}x^2 + \frac{8727639650}{313^4}x + 1$
331	$x^4 + \frac{25305378958}{331^4}x^3 + \frac{4079999534797563}{331^6}x^2 + \frac{25305378958}{331^4}x + 1$
337	$x^4 + \frac{49565720050}{337^4}x^3 + \frac{8335759733455443}{337^6}x^2 + \frac{49565720050}{337^4}x + 1$
347	$x^4 + \frac{1212080056}{347^4}x^3 - \frac{3187187796793566}{347^6}x^2 + \frac{1212080056}{347^4}x + 1$
373	$x^4 + \frac{36459389450}{373^4}x^3 + \frac{5050964299804611}{373^6}x^2 + \frac{36459389450}{373^4}x + 1$
379	$x^4 - \frac{7368210626}{379^4}x^3 + \frac{2095480970724651}{379^6}x^2 - \frac{7368210626}{379^4}x + 1$
383	$x^4 + \frac{921487336}{383^4}x^3 - \frac{5832545849638446}{383^6}x^2 + \frac{921487336}{383^4}x + 1$
389	$x^4 + \frac{1158906320}{389^4}x^3 - \frac{5962663312579758}{389^6}x^2 + \frac{1158906320}{389^4}x + 1$
397	$x^4 - \frac{11841943100}{397^4}x^3 - \frac{3504881162458794}{397^6}x^2 - \frac{11841943100}{397^4}x + 1$
401	$x^4 + \frac{18202354720}{401^4}x^3 - \frac{957643871904318}{401^6}x^2 + \frac{18202354720}{401^4}x + 1$
563	$x^4 - \frac{160493053336}{563^4}x^3 + \frac{83627470409245314}{563^6}x^2 - \frac{160493053336}{563^4}x + 1$
569	$x^4 - \frac{18127289440}{569^4}x^3 - \frac{52094367893657118}{569^6}x^2 - \frac{18127289440}{569^4}x + 1$
571	$x^4 - \frac{55075643282}{571^4}x^3 + \frac{26754041858593083}{571^6}x^2 - \frac{55075643282}{571^4}x + 1$
577	$x^4 - \frac{384554858350}{577^4}x^3 + \frac{183058049445541971}{577^6}x^2 - \frac{384554858350}{577^4}x + 1$
587	$x^4 - \frac{279706107464}{587^4}x^3 + \frac{123214314538450914}{587^6}x^2 - \frac{279706107464}{587^4}x + 1$
613	$x^4 - \frac{77232825670}{613^4}x^3 + \frac{36496351630496211}{613^6}x^2 - \frac{77232825670}{613^4}x + 1$
619	$x^4 - \frac{438694934158}{619^4}x^3 + \frac{233408054712830523}{619^6}x^2 - \frac{438694934158}{619^4}x + 1$
631	$x^4 + \frac{158962935334}{631^4}x^3 + \frac{1718328624127611}{631^6}x^2 + \frac{158962935334}{631^4}x + 1$

641	$x^4 + \frac{488267093600}{641^4}x^3 + \frac{263991060797310402}{641^6}x^2 + \frac{488267093600}{641^4}x + 1$
647	$x^4 + \frac{511190451976}{647^4}x^3 + \frac{299366951978338674}{647^6}x^2 + \frac{511190451976}{647^4}x + 1$
673	$x^4 - \frac{13563858770}{673^4}x^3 + \frac{162061572362830611}{673^6}x^2 - \frac{13563858770}{673^4}x + 1$
683	$x^4 - \frac{66563266936}{683^4}x^3 - \frac{121423072365311646}{683^6}x^2 - \frac{66563266936}{683^4}x + 1$
691	$x^4 - \frac{549274657436}{691^4}x^3 + \frac{352003029734324886}{691^6}x^2 - \frac{549274657436}{691^4}x + 1$
701	$x^4 + \frac{564631759120}{701^4}x^3 + \frac{350990828251510002}{701^6}x^2 + \frac{564631759120}{701^4}x + 1$
733	$x^4 - \frac{735606056900}{733^4}x^3 + \frac{504550137214879638}{733^6}x^2 - \frac{735606056900}{733^4}x + 1$
739	$x^4 - \frac{891640147684}{739^4}x^3 + \frac{652008527391042006}{739^6}x^2 - \frac{891640147684}{739^4}x + 1$
743	$x^4 + \frac{546613589896}{743^4}x^3 + \frac{353290314292721394}{743^6}x^2 + \frac{546613589896}{743^4}x + 1$
751	$x^4 + \frac{96742225082}{751^4}x^3 - \frac{56525728851331797}{751^6}x^2 + \frac{96742225082}{751^4}x + 1$
757	$x^4 - \frac{242701250890}{757^4}x^3 - \frac{97575349088959869}{757^6}x^2 - \frac{242701250890}{757^4}x + 1$
761	$x^4 - \frac{210785930080}{761^4}x^3 + \frac{389858037163348002}{761^6}x^2 - \frac{210785930080}{761^4}x + 1$
809	$x^4 - \frac{1199237879840}{809^4}x^3 + \frac{1014603715639723362}{809^6}x^2 - \frac{1199237879840}{809^4}x + 1$
811	$x^4 - \frac{1314513842972}{811^4}x^3 + \frac{1176379723629896118}{811^6}x^2 - \frac{1314513842972}{811^4}x + 1$
821	$x^4 - \frac{57575596400}{821^4}x^3 - \frac{405803690171181678}{821^6}x^2 - \frac{57575596400}{821^4}x + 1$
827	$x^4 - \frac{426172609544}{827^4}x^3 + \frac{37630959249069474}{827^6}x^2 - \frac{426172609544}{827^4}x + 1$
853	$x^4 + \frac{397324311670}{853^4}x^3 + \frac{34410814591514883}{853^6}x^2 + \frac{397324311670}{853^4}x + 1$
859	$x^4 + \frac{731459396158}{859^4}x^3 + \frac{725878227966315243}{859^6}x^2 + \frac{731459396158}{859^4}x + 1$
863	$x^4 + \frac{967905524984}{863^4}x^3 + \frac{1008408109732184274}{863^6}x^2 + \frac{967905524984}{863^4}x + 1$
877	$x^4 - \frac{791635955110}{877^4}x^3 + \frac{1064491607489740611}{877^6}x^2 - \frac{791635955110}{877^4}x + 1$
881	$x^4 - \frac{599351209600}{881^4}x^3 + \frac{779162552176253442}{881^6}x^2 - \frac{599351209600}{881^4}x + 1$
887	$x^4 + \frac{831742406264}{887^4}x^3 + \frac{345640264207012914}{887^6}x^2 + \frac{831742406264}{887^4}x + 1$
919	$x^4 - \frac{954197719772}{919^4}x^3 + \frac{544170868711032678}{919^6}x^2 - \frac{954197719772}{919^4}x + 1$
929	$x^4 + \frac{823620936160}{929^4}x^3 + \frac{1097340726097664322}{929^6}x^2 + \frac{823620936160}{929^4}x + 1$
937	$x^4 + \frac{1945025849470}{937^4}x^3 + \frac{2385048755126011731}{937^6}x^2 + \frac{1945025849470}{937^4}x + 1$

941	$x^4 - \frac{179428189840}{941^4}x^3 + \frac{934504705643165682}{941^6}x^2 - \frac{179428189840}{941^4}x + 1$
947	$x^4 - \frac{604673512696}{947^4}x^3 + \frac{1287219622753875714}{947^6}x^2 - \frac{604673512696}{947^4}x + 1$
983	$x^4 + \frac{510124363064}{983^4}x^3 + \frac{1497213212956568754}{983^6}x^2 + \frac{510124363064}{983^4}x + 1$
991	$x^4 - \frac{3379368931798}{991^4}x^3 + \frac{4798660891662172923}{991^6}x^2 - \frac{3379368931798}{991^4}x + 1$
997	$x^4 + \frac{1543373344700}{997^4}x^3 + \frac{2244134916298495158}{997^6}x^2 + \frac{1543373344700}{997^4}x + 1$
1033	$x^4 + \frac{1387073257570}{1033^4}x^3 + \frac{1683610590642402291}{1033^6}x^2 + \frac{1387073257570}{1033^4}x + 1$
1039	$x^4 - \frac{2360589166294}{1039^4}x^3 + \frac{2608749046240065051}{1039^6}x^2 - \frac{2360589166294}{1039^4}x + 1$
1049	$x^4 + \frac{1513812002560}{1049^4}x^3 + \frac{1570134423148892322}{1049^6}x^2 + \frac{1513812002560}{1049^4}x + 1$
1051	$x^4 + \frac{3542823942812}{1051^4}x^3 + \frac{5312603627732750838}{1051^6}x^2 + \frac{3542823942812}{1051^4}x + 1$
1061	$x^4 + \frac{4101342499760}{1061^4}x^3 + \frac{6446854952724637842}{1061^6}x^2 + \frac{4101342499760}{1061^4}x + 1$
1093	$x^4 + \frac{4154833913020}{1093^4}x^3 + \frac{6683509183193866806}{1093^6}x^2 + \frac{4154833913020}{1093^4}x + 1$
1103	$x^4 + \frac{4076792079496}{1103^4}x^3 + \frac{7009921814916160914}{1103^6}x^2 + \frac{4076792079496}{1103^4}x + 1$
1109	$x^4 - \frac{5410819185200}{1109^4}x^3 + \frac{9598415251858344402}{1109^6}x^2 - \frac{5410819185200}{1109^4}x + 1$
1117	$x^4 - \frac{3295938213590}{1117^4}x^3 + \frac{5026781226884334963}{1117^6}x^2 - \frac{3295938213590}{1117^4}x + 1$
1153	$x^4 + \frac{991409030050}{1153^4}x^3 + \frac{4496801677352181411}{1153^6}x^2 + \frac{991409030050}{1153^4}x + 1$
1163	$x^4 + \frac{140761750184}{1163^4}x^3 + \frac{3001039650939026274}{1163^6}x^2 + \frac{140761750184}{1163^4}x + 1$
1171	$x^4 - \frac{5543735671922}{1171^4}x^3 + \frac{10348821127599609003}{1171^6}x^2 - \frac{5543735671922}{1171^4}x + 1$
1181	$x^4 + \frac{146791065520}{1181^4}x^3 + \frac{985280358492442482}{1181^6}x^2 + \frac{146791065520}{1181^4}x + 1$
1187	$x^4 + \frac{6749131664264}{1187^4}x^3 + \frac{13623609252624037314}{1187^6}x^2 + \frac{6749131664264}{1187^4}x + 1$
1213	$x^4 - \frac{4370727949910}{1213^4}x^3 + \frac{7699010627101587891}{1213^6}x^2 - \frac{4370727949910}{1213^4}x + 1$

Example 3. For $D = 316$, $\ell = 3$.

Table 3: Minimal polynomial for $D = 316$, $\ell = 3$

p	P
11	$x^6 + \left(-\frac{4011625559156}{11^{13}}d_K + \frac{9454947156017920}{11^{15}}\right)x^5$ $+ \left(-\frac{973424919872269312}{11^{18}}d_K + \frac{30566221523909604971}{11^{18}}\right)x^4$ $+ \left(-\frac{2161187695579389872872}{11^{21}}d_K + \frac{328033382363425721856}{11^{19}}\right)x^3 + \dots$
17	$x^6 + \left(-\frac{26786092180976848}{17^{15}}d_K + \frac{6214249541256688496}{17^{15}}\right)x^5$ $+ \left(\frac{1372430661881939252480}{17^{18}}d_K + \frac{31830661833616258412003}{17^{18}}\right)x^4$ $+ \left(\frac{11184476509887299295026784}{17^{21}}d_K + \frac{86998380108866831131017952}{17^{21}}\right)x^3 + \dots$
19	$x^6 + \left(\frac{979994338165336384}{19^{15}}d_K - \frac{862232942990910782}{19^{14}}\right)x^5$ $+ \left(-\frac{7564798306461905365120}{19^{18}}d_K + \frac{303273139610933415761687}{19^{18}}\right)x^4$ $+ \left(\frac{5551412252309566348937088}{19^{20}}d_K - \frac{1434096286354822195394363884}{19^{21}}\right)x^3 + \dots$
23	$x^6 + \left(\frac{36373134929040298640}{23^{15}}d_K + \frac{724951328582380894208}{23^{15}}\right)x^5$ $+ \left(\frac{1150744294065705142550528}{23^{18}}d_K + \frac{17450485689451075423414451}{23^{18}}\right)x^4$ $+ \left(\frac{17346839025891502138196020000}{23^{21}}d_K + \frac{12469006809858363185330245632}{23^{20}}\right)x^3 + \dots$
29	$x^6 + \left(\frac{964221973579855949552}{29^{15}}d_K - \frac{7133582441830265908936}{29^{15}}\right)x^5$ $+ \left(-\frac{14055483301694683892315008}{29^{18}}d_K + \frac{508846783339452434354993627}{29^{18}}\right)x^4$ $+ \left(\frac{781725913142232743483434782432}{29^{21}}d_K - \frac{4453703961410409715859810903312}{29^{21}}\right)x^3 + \dots$
31	$x^6 + \left(\frac{370483245538566861952}{31^{15}}d_K + \frac{1888124959146159771646}{31^{14}}\right)x^5$ $+ \left(-\frac{21513160385261721990797056}{31^{18}}d_K + \frac{2824260083651763715895378495}{31^{18}}\right)x^4$ $+ \left(-\frac{26506590395851482765433481472}{31^{20}}d_K + \frac{85247565574291880392519363545980}{31^{21}}\right)x^3 \dots$
37	$x^6 + \left(\frac{18277117390339311469184}{37^{15}}d_K - \frac{1043803678832859960854710}{37^{15}}\right)x^5$ $+ \left(-\frac{87518458130490828099291904}{37^{17}}d_K + \frac{69880866667234096869978439463}{37^{18}}\right)x^4$ $+ \left(\frac{234362548456181589810575609523456}{37^{21}}d_K - \frac{3431762378964407230453108457846708}{37^{21}}\right)x^3 \dots 1$
41	$x^6 + \left(\frac{110377903420949287351168}{41^{15}}d_K + \frac{1581147378101471487235504}{41^{15}}\right)x^5$ $+ \left(\frac{9505662343642108158681221120}{41^{18}}d_K + \frac{203810126618450945008945474739}{41^{18}}\right)x^4$

	$+ (\frac{1322810427963749968053553606392576}{41^{21}} d_K + \frac{8950118998450578484328757975516512}{41^{21}}) x^3 \dots$
83	$x^6 + (\frac{6839657750570247695919117940}{83^{15}} d_K + \frac{164673399831810810544239213952}{83^{15}}) x^5$ $+ (\frac{10451134756408149341097576797715968}{83^{18}} d_K + \frac{161851249174932634819277921018454971}{83^{18}}) x^4$ $+ (\frac{7701247825801736457173666377156542674600}{83^{21}} d_K + \frac{113261541581754096383580589122599461626624}{83^{21}}) x^3 + \dots$
109	$x^6 + (\frac{206633357475808612231028363264}{109^{15}} d_K - \frac{915774864648046091726227350226}{109^{15}}) x^5$ $+ (\frac{24864498327815852097007938002345984}{109^{18}} d_K + \frac{1139781986662190813170557661420669223}{109^{18}}) x^4$ $+ (\frac{35231406356336423531991229313855650996224}{109^{21}} d_K + \frac{3132994895585745007303474766893296277585060}{109^{21}}) x^3 + \dots$
113	$x^6 + (\frac{492002011444443392371566142592}{113^{15}} d_K + \frac{3547249858219719645960762068192}{113^{15}}) x^5$ $+ (-\frac{324397521344284837186538396435697664}{113^{18}} d_K + \frac{21155257500432313933767880104277992227}{113^{18}}) x^4$ $+ (\frac{1132595330439977138487829789420488985060608}{113^{21}} d_K - \frac{904169273416751475915418998715588020276800}{113^{21}}) x^3 + \dots$
131	$x^6 + (-\frac{3319176420810989632124579456404}{131^{15}} d_K - \frac{9907774195903023964383738284032}{131^{15}}) x^5$ $+ (-\frac{640860190533654903191322616678309888}{131^{18}} d_K + \frac{154298514318712657489662343372638010907}{131^{18}}) x^4$ $+ (-\frac{23614655300746992387211469252061461320105448}{131^{21}} d_K + \frac{227137410679204091009329997239308834177036288}{131^{21}}) x^3 + \dots$
137	$x^6 + (-\frac{7494153686346544726858204840064}{137^{15}} d_K - \frac{294247034771690392115057967817712}{137^{15}}) x^5$ $+ (\frac{59605680002589942105214621950122326016}{137^{18}} d_K + \frac{1166925527877180446240826392058497685875}{137^{18}}) x^4$ $+ (-\frac{203316083378735539703057203749861917970772224}{137^{21}} d_K - \frac{3441964363954266411685051599315236355399117280}{137^{21}}) x^3 + \dots$
149	$x^6 + (\frac{89029302724608158239592245000064}{149^{15}} d_K - \frac{77734837024827096800506908356440}{149^{15}}) x^5$ $+ (-\frac{27978434607715341832242963236183630848}{149^{18}} d_K + \frac{10711045197607480594786724392351400503211}{149^{18}}) x^4$ $+ (\frac{2477119150104010149292115616503505470814948096}{149^{21}} d_K - \frac{2386873982304380536982549077504011869920019504}{149^{21}}) x^3 + \dots$
151	$x^6 + (-\frac{95439231688984541347070914454528}{151^{15}} d_K + \frac{271562102333996859461781356502166}{151^{15}}) x^5$ $+ (-\frac{222903756629363711197400316112748640256}{151^{18}} d_K + \frac{8963592663508684735825019984988367645055}{151^{18}}) x^4$ $+ (-\frac{1853822990904032197427928484899246777051850752}{151^{21}} d_K + \frac{20833614453664678652394651337469508541265458100}{151^{21}}) x^3 + \dots$
157	$x^6 + (\frac{38310337015811520223021739577856}{157^{15}} d_K - \frac{2605004762902410857600236526729138}{157^{15}}) x^5$ $+ (-\frac{354024603239036018647239562211636319232}{157^{18}} d_K + \frac{17308720768570085099627856248043497432903}{157^{18}}) x^4$ $+ (\frac{1647716665194940746496797098500891243500401664}{157^{21}} d_K - \frac{80931920924366204796222910688546404839905869084}{157^{21}}) x^3 + \dots$
163	$x^6 + (-\frac{41244833968689576252262129639424}{163^{15}} d_K - \frac{1339794167166683396004916825803758}{163^{15}}) x^5$

	$+(-\frac{3026861359124837645338867091149176832}{163^{18}}d_K - \frac{6196008571536838926234729189345249904057}{163^{18}})x^4$ $+(\frac{1592336552168002673293618595257997627975122944}{163^{21}}d_K + \frac{50632799460970461147628668735596146035461201756}{163^{21}})x^3 + \dots$
167	$x^6 + (\frac{617803647973443872413404118455832}{167^{15}}d_K - \frac{19335787071880500091300563143936}{167^{15}})x^5$ $+(\frac{9267365421822534400120766630951806976}{167^{18}}d_K + \frac{114751245489436670835910418571387010411475}{167^{18}})x^4$ $+(\frac{38671567334000649914502319287188746620735356720}{167^{21}}d_K + \frac{2304857387580431934579371055062966582004739584}{167^{21}})x^3 \dots$
173	$x^6 + (-\frac{717729270186101490770266045840256}{173^{15}}d_K + \frac{3398736957323970552023921817924952}{173^{15}})x^5$ $+(-\frac{2434986127799210981668227063732281918464}{173^{18}}d_K + \frac{130251006431207592345576757112385790517435}{173^{18}})x^4$ $+(-\frac{46459607795973930270320386191902193231127831296}{173^{21}}d_K + \frac{298296365790642040751238138356817807018288104240}{173^{21}})x^3 + \dots$
179	$x^6 + (-\frac{290942753025673994919117497518300}{179^{15}}d_K - \frac{2501646311697699858150161795723488}{179^{15}})x^5$ $+(\frac{13449021848929487889064478633192179072}{179^{17}}d_K + \frac{52686071790610196403743036640886817300987}{179^{18}})x^4$ $+(-\frac{9561076921554691497700699913493594359841044216}{179^{21}}d_K - \frac{224525295204358471620115395902309931308901071808}{179^{21}})x^3 \dots$
193	$x^6 + (-\frac{453495377447032449882717104896}{193^{15}}d_K + \frac{6792593548333929673430854959178178}{193^{15}})x^5$ $+(\frac{2334209941898085358862628689299692251648}{193^{18}}d_K + \frac{10272628123889104307338578487675206889343}{193^{18}})x^4$ $+(\frac{39440430766217968877591746500620879895666633216}{193^{21}}d_K - \frac{941519561397991532252102719902307445445867617156}{193^{21}})x^3 + \dots$
197	$x^6 + (\frac{3606142870734732678589029715161088}{197^{15}}d_K + \frac{9270969254504053500328480401686152}{197^{15}})x^5$ $+(-\frac{4266172795650567723742694210056053882880}{197^{18}}d_K + \frac{587572894690848591529569989798424821058251}{197^{18}})x^4$ $+(\frac{240411546264149480320552779469213597493612371968}{197^{21}}d_K - \frac{1980024969084545173405104899291660518087962246256}{197^{21}})x^3 + \dots$
223	$x^6 + (-\frac{23797940519429919319732298703345664}{223^{15}}d_K + \frac{110920786398292550542333116000070142}{223^{15}})x^5$ $+(-\frac{344159302075702406198987771011337669496832}{223^{18}}d_K + \frac{4296297987701498650844884379097668755610303}{223^{18}})x^4$ $+(-\frac{1813201160055871681929975963497976944738644340736}{223^{21}}d_K + \frac{108163640162860658233581420713277022682942435895556}{223^{21}})x^3 + \dots$